R20

Code No.: 20ITC20

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. (IT, AI&DS) V Sem (Main) Examination Dec 2022 / Jan 2023

Computer Networks

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A & Part –B (Internal Choice)** at one place in the same order

Part - A
(5Q X 3M = 15 Marks)

			M	CO	BT
1		Compare OSI and TCP/IP Reference Models.	(3)	1	4
2		State about Tunneling concept.	(3)	2	2
3		Give a brief notes on Multiplexing in transport layer.	(3)	3	2
4		How does leaky bucket algorithm improve the Quality of service in TCP?	(3)	4	3
5		Write short notes on WWW.	(3)	5	2

Part – B
(5Q X 9M = 45 Marks)

			M	CO	BT
6	(a)	What are the different layers in the OSI Reference Model? Explain the functions of each layer.	(5)	1	2
	(b)	Discuss about collision free protocols with suitable examples.	(4)	1	3
		(OR)			
7	(a)	Explain the three persistence methods of the CSMA?	(5)	1	2
	(b)	Consider a 10 Mbps Ethernet LAN that has stations attached to a 2.5 km long coaxial cable. Given that the transmission speed is 2.3 x 108 m/sec, the packet size is 128 bytes out of which 30 bytes are overhead, find the effective transmission rate and maximum rate/throughput at which the network can send data.	(4)	1	3
8	(a)	Explain Distance Vector Routing algorithm with an example.	(5)	2	3
	(b)	Elaborate the concept of Packet Fragmentation.	(4)	2	2
		(OR)			
9	(a)	Two computers C1 and C2 are configured as follows. C1 has IP address 203.197.2.53 and netmask 255.255.128.0. C2 has IP address 203.197.75.201 and netmask 255.255.192.0. Find whether they belong to same network or different networks?	(5)	2	3
	(b)	Describe Link State Routing algorithm with an example.	(4)	2	2
10	(a)	Explain the mechanism in transport layer for controlling congestion?	(5)	3	3
	(b)	Compare the differences between flow-control, error control and congestion control.	(4)	3	4
		(OR)			
11	(a)	How three way handshake ensures reliability of TCP.	(5)	3	3
	(b)	Illustrate the concept of Crash Recovery in Transport Layer.	(4)	3	2

12	(a)	Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.	(5)	4	3
	(b)	Explain Sliding Window Management in TCP.	(4)	4	2
		(OR)			
13	(a)	Explain various steps involved in releasing a TCP connection.	(5)	4	3
	(b)	Draw the TCP segment header format and explain each of its fields.	(4)	4	2
14	(a)	Discuss on Domain Name System in detail.	(5)	5	2
	(b)	How does the user gets E-mail from the message transfer agent?	(4)	5	2
		(OR)			
15	(a)	Describe the concept of streaming stored media.	(5)	5	3
	(b)	Outline the concept of Content Delivery Networks in detail.	(4)	5	2

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